Comparison of Lead acid and Li-ion Battery in Solar Home System of Bangladesh

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Abstract—**Different technical issues regarding the use of Li-ion and lead acid battery in solar photovoltaic system is presented in this study. Economic comparison for using these two types of batteries in solar home system (SHS) of Bangladesh is also presented in terms of net present cost (NPC). Li-ion batteries, especially LiFeSO4 batteries, are technically more advantageous for use in intermittent solar photovoltaic system than conventional lead acid battery. Fast charging rate, high energy density, extended cycle life, low maintenance are advantages of Li-ion batteries compared to lead acid batteries. At present main disadvantage of Li-ion batteries is its high capital cost. From HOMER (an micro-grid simulation software) it is found that instead of high cycle life and maximum depth of discharge (DOD) of Li-ion batteries than lead acid ones, NPC of SHS with Li-ion battery is higher than that of system with lead acid battery if project lifetime is less than 5 years. For project lifetime greater than 5 years, NPC of Li-ion based system is comparable to lead acid system. Moreover, for small size of systems using Li-ion batteries in SHS is economical than lead acid batteries. As system size increases, use of Li-ion battery in SHS becomes costlier than leas acid battery. Environmental effect of two considered types of batteries is also discussed here.**

Keywords-component; Lead acid battery; Li-ion battery; Solar Home System (SHS);Economic comparison; Technical comparison

I. INTRODUCTION

In Bangladesh, at present, grid electricity is not sufficient to meet the electricity demand of country. Government of Bangladesh declares in its vision to bring the whole country under electricity by 2021 [1]. Mainly rural people are deprived of electricity. To electrify rural people and booster economic development, a non-government organization (NGO) IDCOL started solar project named Solar Home System (SHS) in 2003. In this project solar panels of different capacities are being installed according to consumers need. The project was initially funded by World Bank and Global Environment Facility (GEF). Now GIZ, KfW, ADB, IDB, GPOBA, JICA, USAID and DFID are supporting to expand the SHS project. At present 47 other organizations, called Partner Organization (PO), are collaborating IDCOL in SHS project. About 3 million SHSs have already been installed by 2014 [2]. Installed SHS of different capacities by IDCOL is shown in figure 1. In SHS, a battery bank is charged from connected solar panel during daytime and this battery bank supply electricity to load during night. Typical load used in this system are mainly DC

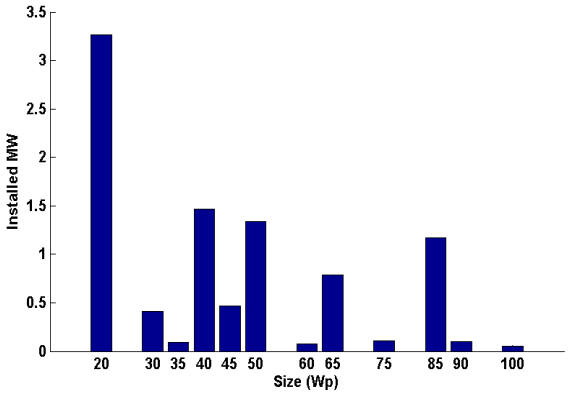


Figure 1. Installed MW of different size of systems by 2014

light, black-white TV, DC fan and cell phone. Battery used in SHSs is mainly lead acid battery. At present, there are 16 domestic battery suppliers in IDCOL SHS program [3].

Due to high energy density Li-ion batteries have become an attractive option for use in solar-PV system. Use of $LiFeSO_4$ battery in standalone solar system is discussed in study [4]. Optimum charging of Li-ion battery from solar PV panel is analyzed in study [5]. In study [6] authors compare the economic aspects between Li-ion batteries and lead acid batteries in solar system. IDCOL also plans to introduce Li-ion battery in small size SHSs by 2016. In this study, different technical parameters of considered two types of batteries are analyzed to justify which one is advantageous for use in solar PV system. Secondly an economic analysis is made to determine the feasibility of Li-ion battery based SHS over existing lead acid battery based SHS in Bangladesh.

II. COMPARISON OF TECHNICAL PARAMETERS

A. Charging time

It is required about 2-3 hours for Li-ion battery to charge fully while for lead acid battery this time is nearly 12-16 hours [7,13]. Figure 2 shows the charging characteristics of lead acid and Li-ion battery.

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Lead acid battery should be charged in 3 stages which are 1) Constant-current charge 2) topping charge and 3) float charge. After constant charge, when lead-acid battery become full charged, topping charge needs to provide to maintain this full charged state. Li-ion battery does not need to charge fully like lead-acid battery. This is a great advantage for using Li-ion battery in solar PV system. During cloudy and low radiation day, output of PV panel decreases. As lead acid battery requires significantly more time to charge fully than Li-ion battery, decreased PV output may be insufficient to charge lead acid battery. If this scenario continues then the performance of lead acid battery will decrease due to sulfation of electrodes. While lower state of charge (SOC) is not a problem for Li-ion battery. Rather maximum voltage per cell leads to stress in Li-ion battery. So using Li-ion battery is advantageous than lead acid battery in solar PV system.

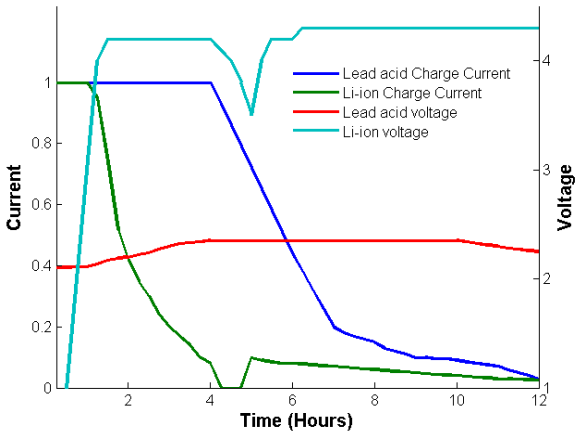


Figure 2. Charging characteristics of lead acid and Li-ion battery (charging current is normalized)

B. *Cycle Life*

Cycle life of typical Absorbent Glass Mat (AGM) type lead acid battery varies from 500-1000 cycles [8]. This cycle life is further lower for flooded lead acid battery. On the other hand cycle life of $LiFeSO_4$ battery varies from 2000 – 5000 cycles. Figure 3 shows the cycle life comparison of lead acid and Li-ion battery. Cycle life of each type of battery is affected by depth of discharge (DOD), discharge rate and temperature but lead acid battery is more sensitive to these factors. As cycle life is affected by DOD, for same size, lead acid battery with 30% DOD can yield same cycle life that of Li-ion battery with 75% DOD. That means lead acid battery must be 2.5 times larger in size to get comparable cycle life.

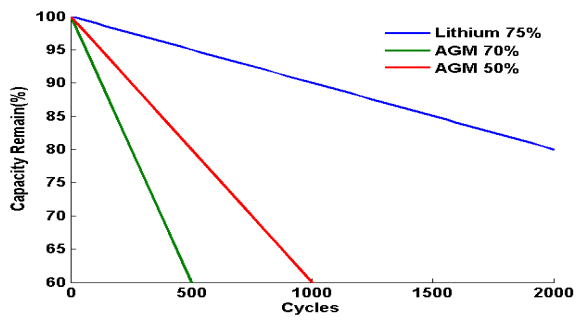


Figure 3. Cycle life comparison of lead acid and Li-ion battery

C. *Maintenance*

The basic chemistry of lead acid battery is

$$Pb(s)+PbSO_4(s)+2H_2SO_4(aq) \rightleftharpoons 2PbSO_4(s)+2H_2O(l) \quad (1)$$

If the cells are overcharged then the physical property of $PbSO_4$ changed and then $PbSO_4$ does not convert to Pb^{2+} and SO_4^{2-} during charging. As a result, specific gravity of electrolyte (H_2SO_4) decreases which hamper the rate of reaction. To avoid this situation it is recommended to charge the lead acid battery cells for prolonged time at a low charge rate.

The specific gravity of the electrolyte also permanently decreases due to aging effect. In SHS mainly flooded lead acid batteries are used. Flooded lead acid batteries require 1) upright orientation to prevent electrolyte loss 2) ventilated environment to diffuse gas and 3) routine maintenance of electrolyte.

The user of lead acid battery needs to periodically topped-off battery with distilled water. While Li-ion batteries are quiet maintenance free. In Li-ion battery it is to ensure that each cell is equally charged and voltage balance exists in battery bank. This feature is maintained automatically by battery management system (BMS) installed within Li-ion battery bank.

D. *Safety*

Both lead-acid and Li-ion battery have safety issues. In lead-acid battery, after battery becomes full charged, O_2 gas form at positive electrode and H_2 gas forms at the negative electrode. This gas mixture is known as detonating gas which exhibits 4-72% explosiveness [10]. For this reason, the charging voltage should be limited and the battery bank room should be adequately ventilated to keep H_2 concentration sufficiently low and thus avoiding explosion.

In SHS, normally flooded lead acid batteries are used. These batteries contain aqueous electrolyte and are operated nearly end-of-charge voltage. So a small amount of gas unavoidable form and gradually water is lost. Continuous overcharge expedite water loss, promote electrode corrosion and reduce battery life span.

On the other hand, Li-ion batteries run safely within operating voltage range but become unstable if charged to higher than specified voltage. Li-ion can't absorb overcharge and charge current must be cut-off when charge fully. Prolonged charging above 4.3 V/cell forms plating of mettalic Li on the anode, while the cathode material becomes an oxidizing agent. This makes the battery unstable by producing CO_2 gas. The CO_2 gas raises cell pressure and if charge continues the current interrupt device is responsible to disconnect the current at specified pressure.

In lead acid battery system, charge controller is used to control charging and discharging of battery bank. In Li-ion battery system, BMS is used to maintain safety specifications of battery system.

E. Recycling

Recycling of Li-ion batteries are expensive than lead acid batteries. Also recycling efficiency of Li-ion battery is only 50% while it is greater than 95% for lead acid battery [12]. In Bangladesh, at present, Rahim Afrooz, Panna Battery and HAMKO have independent recycling facility. Other battery suppliers have made formal agreement with these 3 recyclers. As Li-ion battery manufacturing plant is not available at present, there is no Li-ion battery recycling facility in this country.

III. Economic Comparison

A. Capital Cost

At present, capital cost of lead acid battery is far less than Li-ion battery of same Ah size. Deep cycle lead acid battery is dominated in market and used in wide range of applications. On the other hand deep cycle $LiFeSO_4$ batteries are used mainly in emerging electric vehicle (EV) market. Due to high energy density per weight, Li-ion batteries are better option than lead acid batteries in EV application. In solar application, where space is major issue, $LiFeSO_4$ can save up-to 50% space than lead acid while storing same energy. Present capital cost of $LiFeSO_4$ of different rating and different manufacturers are shown in table 3. In Bangladesh, lead acid battery is a well established sector and suppliers of battery are available. In present situation, solar $LiFeSO_4$ batteries can be imported from India or China at relatively lower shipping cost.

In table 3 cost of same size battery from different manufactures are presented. Costs from different manufactures are significantly different due to different technology and variation of battery quality.

B. Net Present Cost (NPC) comparison

Among different sizes of SHSs about 15% systems (in installed MW view) are 50Wp until 2014. The allowable load for 50Wp system according to Grameen-shakti is shown in table 1 [17].

In SHS, battery is charged during daytime from PV panel and discharged during night to supply electricity. For 12 V - 50Wp, system maximum discharge current is calculated as follow

$$P=VI$$
$$I=P/V$$
For 1 LED light: $I_{LED} = 3/12\ A$
For 1 TV: $I_{TV} = 10/12\ A$
For 1 charger: $I_{Charger} = 3/12A$

Here, P is power rating, V is voltage and I is current. So for 50 Wp system,

Maximum discharge current = $3I_{LED} + I_{TV} + I_{Charger} \approx 2$ A

If 2 days autonomy is considered then battery in SHS must be able to supply 2 A current for 4 hours without charging 2 days.

The load profile for 50Wp SHS is shown in figure 4. First lead acid battery is used in system and net present cost (NPC) of system is calculated from HOMER simulation. Then lead acid battery is replaced by $LiFeSO_4$ battery and the NPC of system is calculated. Table 4 shows the considered lead acid and Li-ion battery parameters. Economic parameters considered in simulation are shown in table 2.

NPC of 50Wp SHS with two types of batteries are calculated for different project lifetime. The calculated NPC for SHS with lead acid battery and $LiFeSO_4$ battery is plotted against project lifetime as shown in figure 5. From the figure 5 it is found that $LiFeSO_4$ based system is economical if system lifetime is greater than 5 years.

In figure 5, lifetime of $LiFeSO_4$ is 10 years and maximum DOD is 85%. If lifetime of $LiFeSO_4$ battery is 8 years and maximum DOD is 80% then calculated NPC against project lifetime is shown in figure 6. In figure 6, NPC of lead acid battery system is calculated considering lifetime of lead acid battery before replacement as 5 years and maximum DOD as 55%. From figure 6, it is found that in this case, lead acid battery is better in economy than using $LiFeSO_4$ battery in SHS.

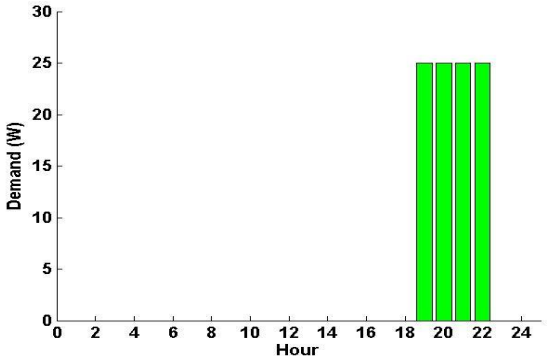


Figure 4. Ideal load profile considered for 50 Wp system

To compare the variation of economy of Li-ion and lead acid based SHS with the variation of system size 20W, 30W, 40W and 65W systems with corresponding load are simulated in HOMER. NPCs of different systems at different project lifetime are calculated and are shown in figure 7. From figure 7 it is found that Li-ion based SHS is economical than that of lead acid based system when system size is small. As system size increases, NPC of Li-ion bassed system exceeds that of lead acid based system.

Table 1. Allowed load for 50Wp system

Load	Nos	Watt Rating
LED Tube light	4	3
15 “ Black-white TV	1	10
Cell phone charger	1	3

Table 2. Economic parameters of considered Lead acid and $LiFeSO_4$ battery for 50Wp system

Parameters	Lead acid battery	Li-ion battery
Capital cost (USD)	50	180
Replacement cost(USD)	50	180
Operation and Maintenance cost (USD)	8	0

Table 3. Capital cost of different sizes of battery

Capacity (Ah)	Li-ion Battery *PL Energy Group Ltd, China(USD)[13]*	Lead acid Battery *Rahim Afrooze Bangladesh [14]*	Li-ion Battery *SMART Battery, USA [15]*	Lead acid battery *Solar Battery [16]*
20	100-150	70.55	290	
30	180-198	76.25		66-70
35			450	73-75
40	232-264	110.85	580	100-115
50	292-330	126.50	690	120-126
60	360-396	133.35		125-129
75		142.20	1000	
80	464-528		1050	132-137

Table 4. Technical parameters of considered Lead acid and $LiFeSO_4$ battery for 50Wp system

Parameters	Lead acid battery	Li-ion battery
Nominal capacity (Ah)	50	30
Nominal voltage (V)	12	12
Minimum state of charge (%)	50	15
Float life (yr)	3	10
Maximum charge current (A)	20	18
Cycle life @ maximum DOD	550	2000

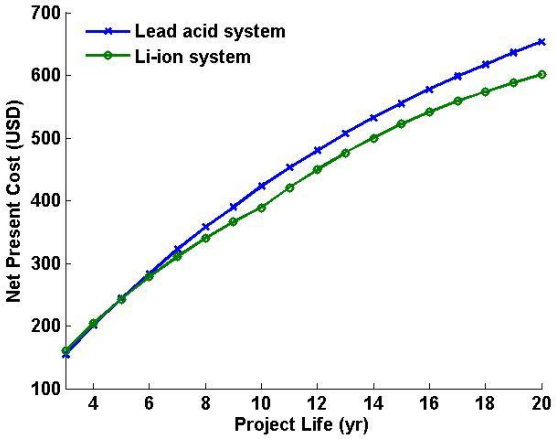


Figure 5. NPC comparison of lead acid and Li-ion based 50Wp system

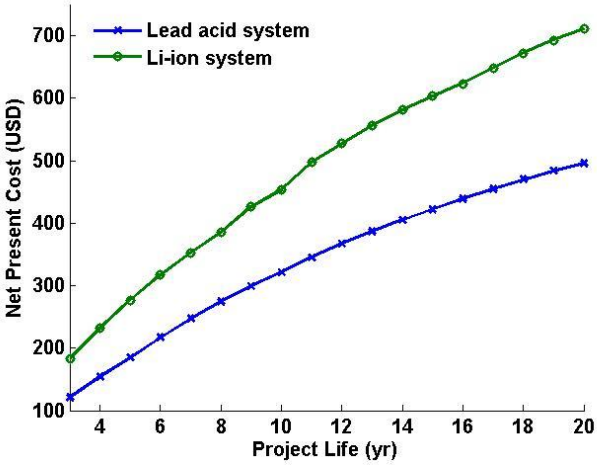


Figure 6. NPC comparison of lead acid and Li-ion based 50Wp system with different battery life cycle

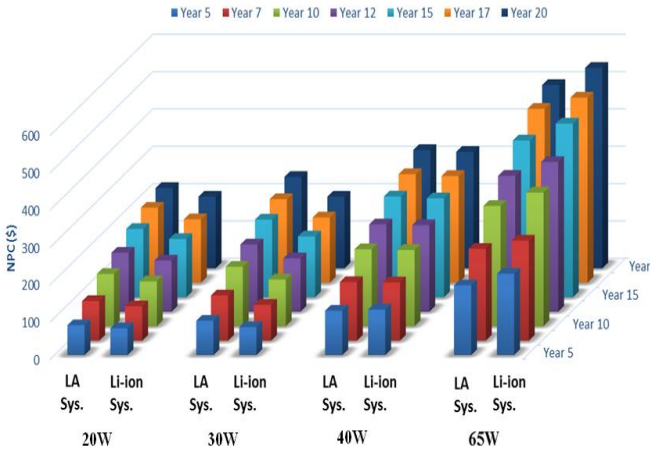


Figure 7. Comparison of NPC variation with system size for lead acid and Li-ion based system

IV. Environmental Effect Comparison

Lead acid batteries contain lead and sulfuric acid solution. Lead is a highly toxic material and sulfuric acid is a corrosive electrolyte. Lead acid batteries should never be disposed of in a solid waste landfill. Contaminates such as lead and sulfuric acid can bleed into soil contaminating vegetation and also affects ground water. If these batteries are disposed of near rivers, streams, lakes or coastal water, the lead and sulfuric acid can also threaten aquatic life.

On the other hand Li-ion batteries contain high level of aluminium, cobalt, copper and lithium. Copper and aluminium are used as current conductors in Li-ion battery and LiCoO2 is used as the cathode material. These materials are not as toxic as lead in lead acid battery. Although in study [18], authors conclude that all Li-ion batteries should be classified as hazardous waste due primarily to excessive level of cobalt and copper and in some case nickel. For these danger associated with battery disposal, the need for recycling becomes more important

V. Conclusion

Due to low maintenance cost, flexibility in SOC, fast charging rate, Li-ion battery, especially $LiFeSO_4$, can be a better replacement of conventional lead acid battery for using

in solar system. Also $LiFeSO_4$ batteries are not environmentally hazardous as lead acid batteries. At present, the main drawback of Li-ion batteries for introducing in SHS is its significantly high price than lead acid batteries. Recently researchers at MIT and a spinoff company called 24M developed an advanced manufacturing approach toward cutting off the cost of Li-ion batteries while improving performance and recycling process [19]. In present situation, lead acid batteries is economically comparable to, even more beneficial than, Li-ion batteries for SHS. However, introducing Li-ion batteries in substitution of lead acid ones in SHS will result in environmental benefits and reduce consumer's maintenance work.

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